

Supporting Information for “The sequestration efficiency of the deep ocean”

Benoît Pasquier^{1,2}, Richard J. Matear³, Matthew A. Chamberlain³, Tilo Ziehn⁴, David K. Hutchinson⁵, François W. Primeau⁶, Yi Liu⁷, and Ann Bardin⁶

¹Independent Researcher

²School of Mathematics and Statistics, University of New South Wales, Sydney, Australia

³Commonwealth Scientific and Industrial Research Organisation, Hobart, Australia

⁴Commonwealth Scientific and Industrial Research Organisation, Aspendale, Australia

⁵Climate Change Research Centre, School of Biological, Earth and Environmental Sciences, University of New South Wales, Sydney, NSW, Australia

⁶Department of Earth System Science, University of California, Irvine, Irvine, USA

⁷Department of Geosciences, High Meadows Environmental Institute, Princeton University, Princeton, USA

Contents of this file

1. Introduction
2. Text Sections S1 to S3
3. Figures S1 to S6

Introduction

This file contains supplementary text and figures for the manuscript “The sequestration efficiency of the deep ocean”. Section S1 provides mathematical details on the transit-time distributions and sequestration efficiency. Section S2 describes the computational methods used to compute the adjoint boundary propagator and the mean time to reemergence. Section S3 details the calibration of the diffusivity constants used to build the ocean transport matrices. Figure S1 illustrates that mean TTD and BIR coincide in periodic flow. Figure S2–S5 compare the transport matrix-computed mean age to the online mean age spun up with Anderson Acceleration. Figure S6 shows the climate change effect on the median and 10th percentile time to reemergence, along with the effect on the sequestration efficiency for time horizons $\tau = 100$ and 300 years, thus complementing main text Fig. 4.

S1. Transit time distribution and sequestration efficiency

S1.1. Tracer Equation, Green Function, and Adjoint Boundary Propagator

To estimate the ocean’s sequestration potential, we consider a passive tracer that is instantly removed when reaching the surface. Its concentration, $c(\mathbf{r}, t)$, can be conveniently approximated by the tracer equation

$$\frac{\partial c}{\partial t} + \mathcal{T}c = -\Omega c/\tau_0, \quad (\text{S1})$$

where $\mathcal{T} \equiv \mathbf{u} \cdot \nabla - \nabla \cdot \mathbf{K} \nabla$ is the advective–eddy-diffusive transport operator (with velocity field $\mathbf{u}$ and diffusivity tensor $\mathbf{K}$). Quick removal over the surface layer with fast timescale $\tau_0 = 1$ s is enforced on the right-hand-side with the mask $\Omega(\mathbf{r}) = 1$ in surface layer and $\Omega(\mathbf{r}) = 0$ elsewhere. (The precise value of τ_0 does not affect our results as long as τ_0 is small relative to the transport timescales.)

The Green function of Eq. (S1), denoted $G(\mathbf{r}, t | \mathbf{r}', t')$, gives the concentration at $(\mathbf{r}, t)$ resulting from a unit impulse source at $(\mathbf{r}', t')$. It is given by the following Green function equation (e.g., Holzer & Hall, 2000):

$$(\partial_t + \mathcal{T} + \Omega/\tau_0) G(\mathbf{r}, t | \mathbf{r}', t') = \delta(t - t') \delta(\mathbf{r} - \mathbf{r}'), \quad (\text{S2})$$

where the right-hand-side is the unit impulse source at $(\mathbf{r}', t')$, with δ being the Dirac delta function. While G theoretically allows us to track the pathways of every molecule of water on its way to the surface, computing G over each possible source and destination time and location is computationally prohibitive. However, we can reduce the dimensionality of this problem by considering many pathways at once in periodic flow through the carefully chosen boundary propagator defined below.

The fraction $dt_f \mathcal{G}^\dagger(\mathbf{r}, t, t_f)$ of water at $(\mathbf{r}, t)$ that will reemerge anywhere into the surface layer during $(t_f, t_f + dt_f)$ is given by (e.g., Pasquier & Holzer, 2016)

$$\mathcal{G}^\dagger(\mathbf{r}, t, t_f) = \int d^3\mathbf{r}' \frac{\Omega(\mathbf{r}')}{\tau_0} G(\mathbf{r}', t_f | \mathbf{r}, t). \quad (\text{S3})$$

The Green function $\mathcal{G}^\dagger$ is called the adjoint boundary propagator (e.g., Haine et al., 2024) and is the solution to the adjoint equation

$$(-\partial_t + \mathcal{M}^\dagger(t)) \mathcal{G}^\dagger(\mathbf{r}, t, t_f) = \frac{\Omega(\mathbf{r}')}{\tau_0} \delta(t_f - t), \quad (\text{S4})$$

where $\mathcal{M}^\dagger(t) \equiv \mathcal{T}^\dagger(t) + \Omega(\mathbf{r})/\tau_0$ and $\mathcal{T}^\dagger = -\mathbf{u} \cdot \nabla - \nabla \cdot \mathbf{K} \nabla$ is the adjoint of $\mathcal{T}$ (note the sign change for advection; e.g., Holzer & Hall, 2000). Equation (S4) can be recast as a final-value problem by integrating in time from $t_f - \epsilon$ to $+\infty$ and taking the $\epsilon \rightarrow 0$ limit.

This gives the equation

$$(-\partial_t + \mathcal{M}^\dagger(t)) \mathcal{G}^\dagger(\mathbf{r}, t, t_f) = 0, \quad (\text{S5})$$

subject to final boundary value $\mathcal{G}^\dagger(\mathbf{r}, t_f, t_f) = \Omega(\mathbf{r})/\tau_0$, which acts as a tracer injection from which water is tracked backwards in time.

S1.2. Sequestration Efficiency

An important property of $\mathcal{G}^\dagger$ for defining the sequestration efficiency is that the function $\tau \mapsto \mathcal{G}^\dagger(\mathbf{r}, t_i, t_i + \tau)$ is the probability density function of the distribution of transit times τ for water at $(\mathbf{r}, t_i)$ to reemerge into the surface at $t_i + \tau$ (the transit time distribution, TTD). The normalization $\int_0^{+\infty} d\tau \mathcal{G}^\dagger(\mathbf{r}, t_i, t_i + \tau) = 1$ follows from the fact that all the water at $(\mathbf{r}, t_i)$ must eventually reemerge into the surface, given the non-zero vertical diffusion (e.g., Holzer & Hall, 2000; Haine et al., 2024). The sequestration efficiency for water at $(\mathbf{r}, t_i)$ for a transit time horizon τ is then defined as the complement of the cumulative distribution function of the TTD:

$$\mathcal{E}(\mathbf{r}, t_i, \tau) \equiv 1 - \underbrace{\int_0^\tau dt \mathcal{G}^\dagger(\mathbf{r}, t_i, t_i + t)}_{\text{fraction reemerging before } \tau} = \underbrace{\int_\tau^\infty dt \mathcal{G}^\dagger(\mathbf{r}, t_i, t_i + t)}_{\text{fraction reemerging after } \tau}. \quad (\text{S6})$$

In practice however, the values of $\mathcal{G}^\dagger$ (and consequently of $\mathcal{E}$) are computed backwards in time from a boundary value at final time t_f (Eq. (S5), illustrated on Fig. S1). As such, the function $\tau \mapsto \mathcal{G}^\dagger(\mathbf{r}, t_f - \tau, t_f)$ is a boundary impulse response (BIR), which is not necessarily a distribution and is distinct from the TTD sought here (Haine et al., 2008). While it is possible to carefully stitch together copies of the BIRs for different final times t_f to obtain the TTDs for different initial times t_i , this is not necessary here for periodic flow when averaging over one periodic year.

S1.3. Mean TTD and BIR are Equal in Periodic Flow

For periodic flow, the annual mean BIR and TTD are identical, their PDF given by:

$$\overline{\mathcal{G}^\dagger}(\mathbf{r}, \tau) \equiv \int_{t_i}^{t_i+1 \text{ yr}} dt \mathcal{G}^\dagger(\mathbf{r}, t, t + \tau) = \int_{t_f}^{t_f+1 \text{ yr}} dt \mathcal{G}^\dagger(\mathbf{r}, t - \tau, t), \quad (\text{S7})$$

where the integrals are over any year (also illustrated in Fig. S1). While others, e.g., Haine et al. (2008), recognized that the expected mean TTD and BIR are identical for statistically periodic (cyclostationary) flow, what is additionally recognized here is that for exactly periodic flow (not just statistically), the mean over any single periodic year of the BIR and TTD are equal. That is, Eq. (S7) is independent of the choice of t_i and t_f . We also emphasize that this equality is exact; the mean BIR is not merely a surrogate for the mean TTD. In other words, taking the mean over one periodic year is enough to “flatten” the contours of $\mathcal{G}^\dagger$, effectively reducing the problem to one of steady flow, for which BIR and TTD are identical (e.g., Haine et al., 2008). In practice, this means that taking the mean of the 12 time series of the BIR over each final month of the periodic year gives the TTD averaged over each initial month directly, without any careful stitching required. The mean sequestration efficiency computed in this work is then given by Eq. (3) in the main text.

S2. Computational Methods

S2.1. Adjoint Boundary Propagator

In practice, for our discretized ocean, we compute the adjoint boundary propagator $\mathcal{G}^\dagger$ by solving the matrix equation

$$(-\partial_t + \mathbf{M}^\dagger(t)) \mathcal{G}^\dagger(t, t_f) = 0 \quad (\text{S8})$$

where $\mathbf{M}^\dagger(t) \equiv \mathbf{V}^{-1} \mathbf{T}(t)^\top \mathbf{V} + \text{diag}(\boldsymbol{\Omega})/\tau_0$ is the matrix representation of $\mathcal{M}^\dagger(t)$ and subject to the final boundary value $\mathcal{G}^\dagger(t_f, t_f) = \boldsymbol{\Omega}/\tau_0$. Applying Euler forward gives a time-stepping scheme that is implicit when going backwards in time, allowing us to accurately compute $\mathcal{G}^\dagger(t - \Delta t, t_f)$ from $\mathcal{G}^\dagger(t, t_f)$ through:

$$(\mathbf{I} + \Delta t \mathbf{M}^\dagger(t - \Delta t)) \mathcal{G}^\dagger(t - \Delta t, t_f) = \mathcal{G}^\dagger(t, t_f). \quad (\text{S9})$$

To compute the BIR for 3000 years, we simulate the adjoint boundary propagator backward in time through repeatedly solving Eq. (S9) month by month. This relatively long time-step of 1 month was chosen for simplicity and efficiency, and does not incur much loss in accuracy because the time-stepping scheme is fully implicit. We thus only need 12 transport matrices, denoted $(\mathbf{T}_m)_{m \in \{1, \dots, 12\}}$, one for each month of the periodic year. We note in passing that the precise time-step size Δt is equal to the mean number of days separating consecutive months and thus varies from month to month in Eq. (S9).

To get the mean TTD, we simply average the 12 BIRs from each final month of the periodic year. This is sufficient because the mean TTD, which is the TTD averaged over one periodic year, is identical to the mean BIR (and given by $\overline{\mathcal{G}^\dagger}$; see Section S1.3 and Fig. S1). That is, we run twelve 3000-year backward simulation of $\mathcal{G}^\dagger(t_f - \tau, t_f)$ for 12 final times t_f taken as the middle of each final month $m \in \{1, \dots, 12\}$. $\overline{\mathcal{G}^\dagger}$ is then given by the average of these 12 simulations weighted by the number of days of each final month (Eq. (S7)).

An important computational advantage for computing the adjoint boundary propagator and the sequestration efficiency using this backwards monthly time-stepping in periodic flow is that it only requires factorizing the 12 matrices $\mathbf{M}_m^\dagger \equiv \mathbf{V}^{-1} \mathbf{T}_m^\top \mathbf{V} + \text{diag}(\boldsymbol{\Omega})/\tau_0$

once and cache these factors. In practice, we used the Pardiso sparse linear solver (Schenk et al., 2001) available from Julia’s LinearSolve.jl package (Kimmerer et al., n.d.) to solve Eq. (S9). An important note for transport matrix users is that in our single-node multi-CPU application the Pardiso solver offered up to $20\times$ speedups for matrix factorizations compared to the default unsymmetric sparse linear “backslash” solver (UMFPACK; Davis, 2004) used in MATLAB and Julia, which is likely because UMFPACK does not parallelize well in comparison to Pardiso (bearing in mind that solver performance can vary strongly with application; Pedroso, 2024). The backward-time simulations of $\mathcal{G}^\dagger$ were computed independently in parallel for each final month, each experiment, each time period, and each ensemble member. The corresponding computations for this work were submitted as single-node 360GB-RAM 24-cores jobs to Australia’s High Performance Cluster, Gadi, taking approximately 30 hours for 3000 simulation years. This represents considerable savings, as it would take roughly 150 times longer to run the ACCESS-ESM1.5 model forward in time for the same 3000 years, bearing in mind that “online” runs would only allow for a limited number of injection locations (there are approximately 50,000 grid cells on the seafloor), whereas the adjoint method employed here provides the TTD for *every* location with a single tracer.

S2.2. Mean Time to Reemergence

The mean time to reemergence averaged over one periodic year, denoted $\overline{\Gamma}^\dagger$, is given by the 1st moment of $\overline{\mathcal{G}}^\dagger$:

$$\overline{\Gamma}^\dagger(\mathbf{r}) = \int_0^\infty d\tau \tau \overline{\mathcal{G}}^\dagger(\mathbf{r}, \tau). \quad (\text{S10})$$

However, one cannot compute $\overline{\mathcal{G}^\dagger}$ over infinitely large times. Instead, we compute the mean time to reemergence, $\Gamma^\dagger(\mathbf{r}, t)$, directly from its tracer equation:

$$\left(-\frac{\partial}{\partial t} + \mathcal{T}^\dagger\right) \Gamma^\dagger(\mathbf{r}, t) = 1 - \frac{\Omega}{\tau_0} \Gamma^\dagger(\mathbf{r}, t). \quad (\text{S11})$$

To compute the periodic solution of Eq. (S11), we use the same periodic equilibrium Newton–Krylov solver as Bardin et al. (2014), albeit in the adjoint context. That is, we apply Newton’s method to find the zero of the function $\mathcal{D}^\dagger(\Gamma^\dagger(t)) \equiv \Gamma^\dagger(t - 1 \text{ yr}) - \Gamma^\dagger(t)$, which gives the projected 1-year drift of $\Gamma^\dagger$, and where the evolution backward in time of $\Gamma^\dagger$ from t to $t - 1$ year is governed by Eq. (S11). Using a Krylov subspace solver allows us to avoid the need to compute the Jacobian of $\mathcal{D}^\dagger$, which would be computationally expensive. Instead, we only need to compute the action of the Jacobian on a vector. In practice, we use the same monthly time-stepping scheme as for the adjoint boundary propagator to evolve $\Gamma^\dagger$ back by one year, which is given in matrix form by

$$(\mathbf{I} + \Delta t \mathbf{M}^\dagger(t - \Delta t)) \mathbf{\Gamma}^\dagger(t - \Delta t) = \mathbf{\Gamma}^\dagger(t) + \Delta t, \quad (\text{S12})$$

where $\mathbf{\Gamma}^\dagger(t)$ (in bold) is the column-vector discretized version of $\Gamma^\dagger(\mathbf{r}, t)$. This is essentially the same as Eq. (S9) but with Δt added to the right-hand side. Given that each monthly backward time step in Eq. (S12) is affine in $\mathbf{\Gamma}^\dagger(t)$, the action of the Jacobian of $\mathcal{D}^\dagger$ on $\mathbf{\Gamma}^\dagger(t)$ is simply computed by applying the time steps of Eq. (S12) but without the constant Δt on the right-hand side. Once the periodic $\mathbf{\Gamma}^\dagger(t)$ has been computed (for, e.g., January), it can be evolved back one last time to get its seasonal variation over the 12 months of the periodic year and averaged over to give $\overline{\mathbf{\Gamma}^\dagger}$.

In practice, we use the Generalized Minimal Residual Algorithm (GMRES; Saad & Schultz, 1986) with the same preconditioner as Bardin et al. (2014), $\mathbf{P} \equiv -\left(\Delta t \overline{\mathbf{M}^\dagger}\right)^{-1} - \mathbf{I}$,

where $\Delta t = 1$ year and $\overline{\mathbf{M}^\dagger}$ is the mean of the 12 monthly $\mathbf{M}_m^\dagger$ matrices. Because $\mathcal{D}^\dagger$ is linear in $\Gamma^\dagger$ here, the periodic solution is found in a single Newton iteration, which as for the adjoint boundary propagator only requires a single factorization of each $\mathbf{M}_m^\dagger$ matrix. Using the same computational resources as for the adjoint boundary propagator, the mean time to reemergence can be computed in less than 30 minutes.

S3. Calibration of Diffusivity Constants

Following Chamberlain et al. (2019), we adjust the diffusivity constants so that the mean age computed with our transport matrices matches the mean age from the online ACCESS-ESM1.5 model, which serves as a reference. But while the mean water age is typically part of climate-model archives, the mean age from the ACCESS-ESM1.5 archives was not spun up long enough to be in equilibrium with the circulation. To avoid running multimillennial simulations to spin up the mean age to equilibrium, we follow Khatiwala (2023, 2024) and use Anderson Acceleration (AA; Anderson, 1965). Below we first define the mean water age, then our matrix-based computation, then how we apply Anderson Acceleration, and finally the calibration process.

The mean age of water, also called the ideal age, $\Gamma(\mathbf{r}, t)$, is given by the tracer equation

$$\frac{\partial}{\partial t} \Gamma + \mathcal{T} \Gamma = 1 - \frac{\Omega}{\tau_0} \Gamma, \quad (\text{S13})$$

where we have omitted the temporal and spatial dependence for brevity. Physically, Γ in Eq. (S13) simply accrues by one second every second and is clamped to zero in the surface layer.

To compute the periodic solution to Eq. (S13), we use the same approach as for the mean time to reemergence, except we apply an Euler backward time-stepping scheme to

retain the stability afforded by a fully implicit scheme. More specifically, we repeatedly solve the matrix equation

$$(\mathbf{I} + \Delta t \mathbf{M}(t + \Delta t)) \mathbf{\Gamma}(t + \Delta t) = \mathbf{\Gamma}(t) + \Delta t, \quad (\text{S14})$$

where $\mathbf{M}(t) \equiv \mathbf{T}(t) + \text{diag}(\mathbf{\Omega})/\tau_0$ is the matrix representation of $\mathcal{M} \equiv \mathcal{T} + \mathbf{\Omega}/\tau_0$. As for the mean reemergence time, the periodic mean age is found using a Newton–Krylov solver that converges to the zero of the function $\mathcal{D}(\mathbf{\Gamma}(t)) \equiv \mathbf{\Gamma}(t + 1 \text{ yr}) - \mathbf{\Gamma}(t)$, which gives the projected 1-year drift of $\mathbf{\Gamma}$.

For spinning up the online age, we repeatedly run the ACCESS-ESM1.5 forward in time over the 1850s decade but replace the mean age at the start of the 1850s with the “accelerated” mean age from the previous AA iteration. Unlike Khatiwala (2023, 2024) who repeated a single year of simulation, we repeat the 1850s decade instead to average out some degree of interannual variability. We save a maximum of $m = 10$ iterates and run AA for 40 iterations, which is equivalent to 400 simulation years and takes roughly 3 weeks of computation time.

To select our preferred diffusivity constants, we repeatedly compute the mean age using our matrix formulation and chose the age with the best skill score, as shown in Fig. S2. The skill-score metric we use is the “simplest” one of Taylor (2001), namely

$$S = \frac{4(1 + R)}{\left(\frac{\sigma_{\text{TM}}}{\sigma} + \frac{\sigma}{\sigma_{\text{TM}}}\right)^2 (1 + R_0)} \quad (\text{S15})$$

where R is the volume-weighted correlation between transport matrix age and AA age, $\sigma = 768$ years is the volume-weighted standard deviation of the AA age, σ_{TM} is the volume-weighted standard deviation of the transport matrix age, and with $R_0 = 1$ as-

sumed as the best correlation attainable. Our preferred diffusivity constant values are $\kappa_V = 3.0 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$, $\kappa_{\text{VML}} = 1.0 \text{ m}^2 \text{ s}^{-1}$, and $\kappa_H = 300 \text{ m}^2 \text{ s}^{-1}$. These values give a skill score $S = 0.997$, a volume-weighted correlation $R = 0.994$, a volume-weighted root-mean-square difference (RMSD) of 94 years, and a mean bias of +37 years. The joint probability density function of the transport matrix age and the AA age is shown in Fig. S3. Basin zonal averages are shown in Fig. S4, and zonally averaged meridional slices are shown in Fig. S5

We note that ACCESS-ESM1.5 appears to have a significantly more sluggish deep ocean circulation compared to the ACCESS1.3 model (compare, e.g., Figs. S2 and S5 herein to Figs. 3 and 4 in the work of Chamberlain et al., 2019). We also note that our diffusivity constants are slightly different from those of Chamberlain et al. (2019). These difference stem from three major differences in the approach used here. The first is that the ACCESS-ESM1.5 model is different from the ACCESS1.3 model. The second is that we build monthly matrices instead of a yearly matrix built from the average of the flow. These monthly matrices capture seasonal mixing that was not captured by Chamberlain et al. (2019). The third is that we used a centered advection scheme, which is much less diffusive than the upwind scheme of Chamberlain et al. (2019). Interestingly, the last two compensate almost perfectly (added mixing from seasonality compensates reduced mixing from the centered scheme).

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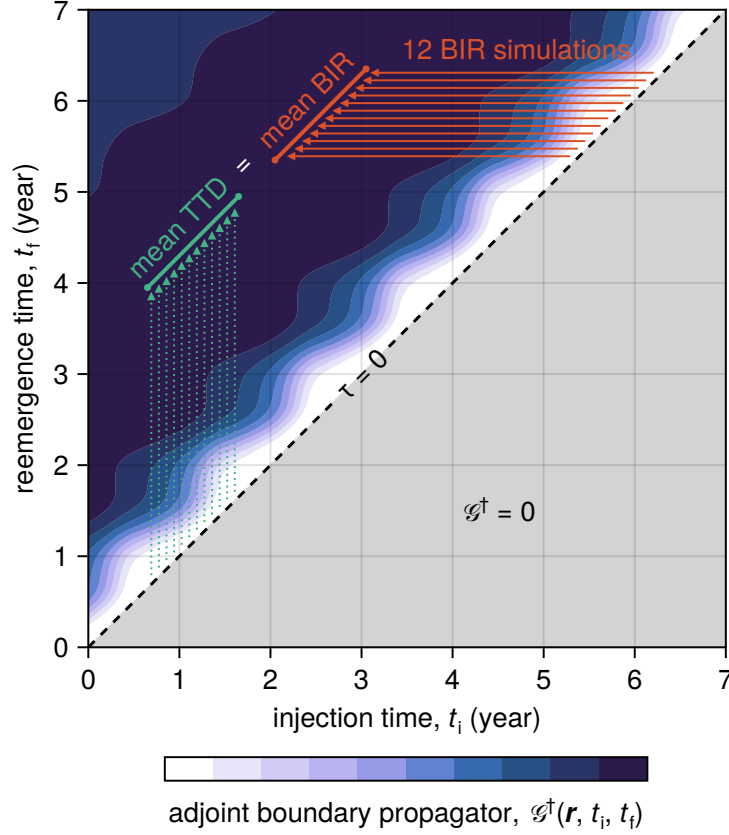


Figure S1. Diagram illustrating the identity between mean transit time distribution (TTD; green) and mean boundary impulse response (BIR; red), overlaid on contours of the adjoint boundary propagator, $\mathcal{G}^\dagger$. (The magnitude and unit of $\mathcal{G}^\dagger$ are omitted for simplicity.) For the same time horizon τ , the TTD and BIR, averaged over any periodic year of initial injection times and final reemergence times, respectively, are identical and given by $\overline{\mathcal{G}^\dagger}(\mathbf{r}, \tau)$ (Eq. (S7)). This identity allows us to compute the mean TTD (and the mean sequestration efficiency, $\overline{\mathcal{E}}$) directly from simply averaging 12 BIR simulations (indicated by solid red arrows) without the need to reconstruct the corresponding 12 TTDs (dashed green arrows).

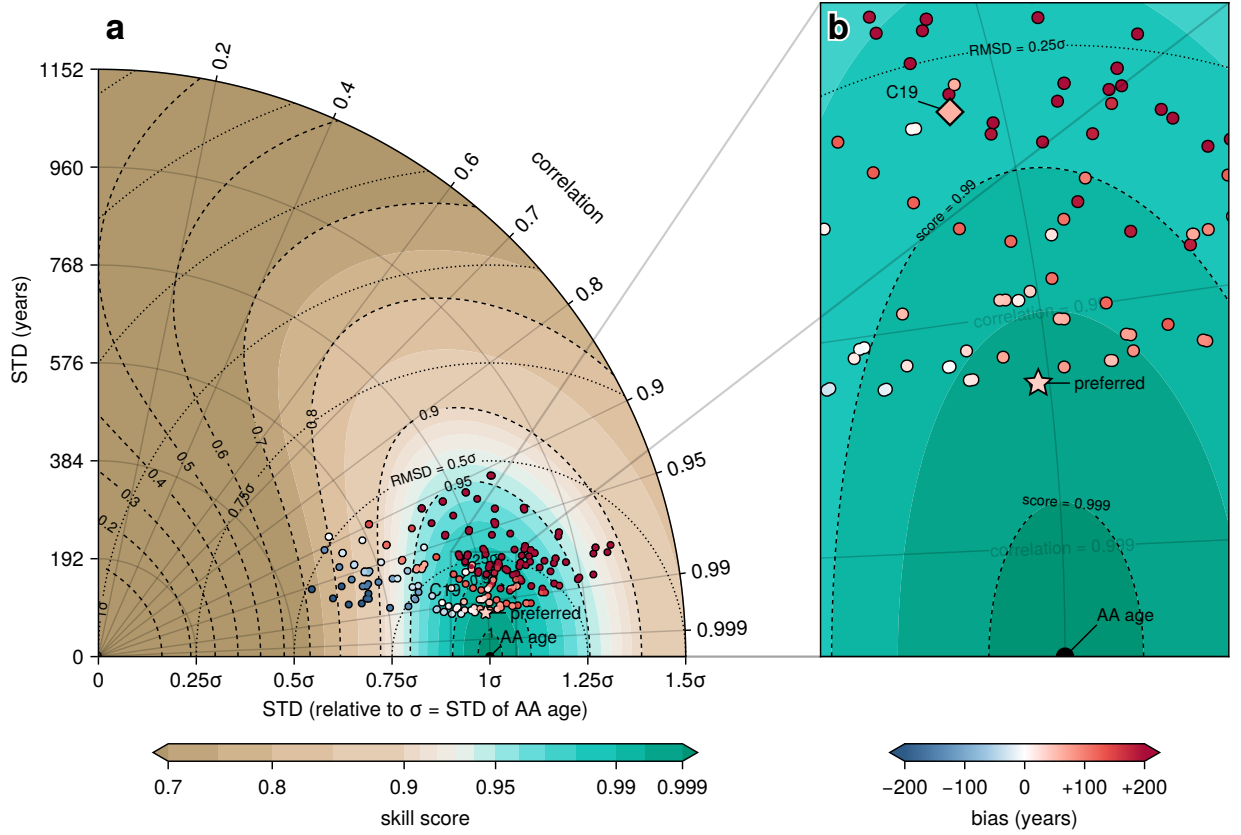


Figure S2. (a) Volume-weighted Taylor diagram (Taylor, 2001) of the transport-matrix solutions for the periodic water age for various diffusivity constants referenced to the forward solution after Anderson Acceleration (AA age). Filled contours represent the “least complicated” skill score (Eq. (4) of Taylor, 2001). The marker colors indicate the volume-weighted mean bias. Our preferred solution is indicated by the star. The solution using our transport matrices (TM) with the diffusivity constants from Chamberlain et al. (2019) is indicated by the diamond and the C19 label. (b) As (a) but zoomed in around our preferred solution.

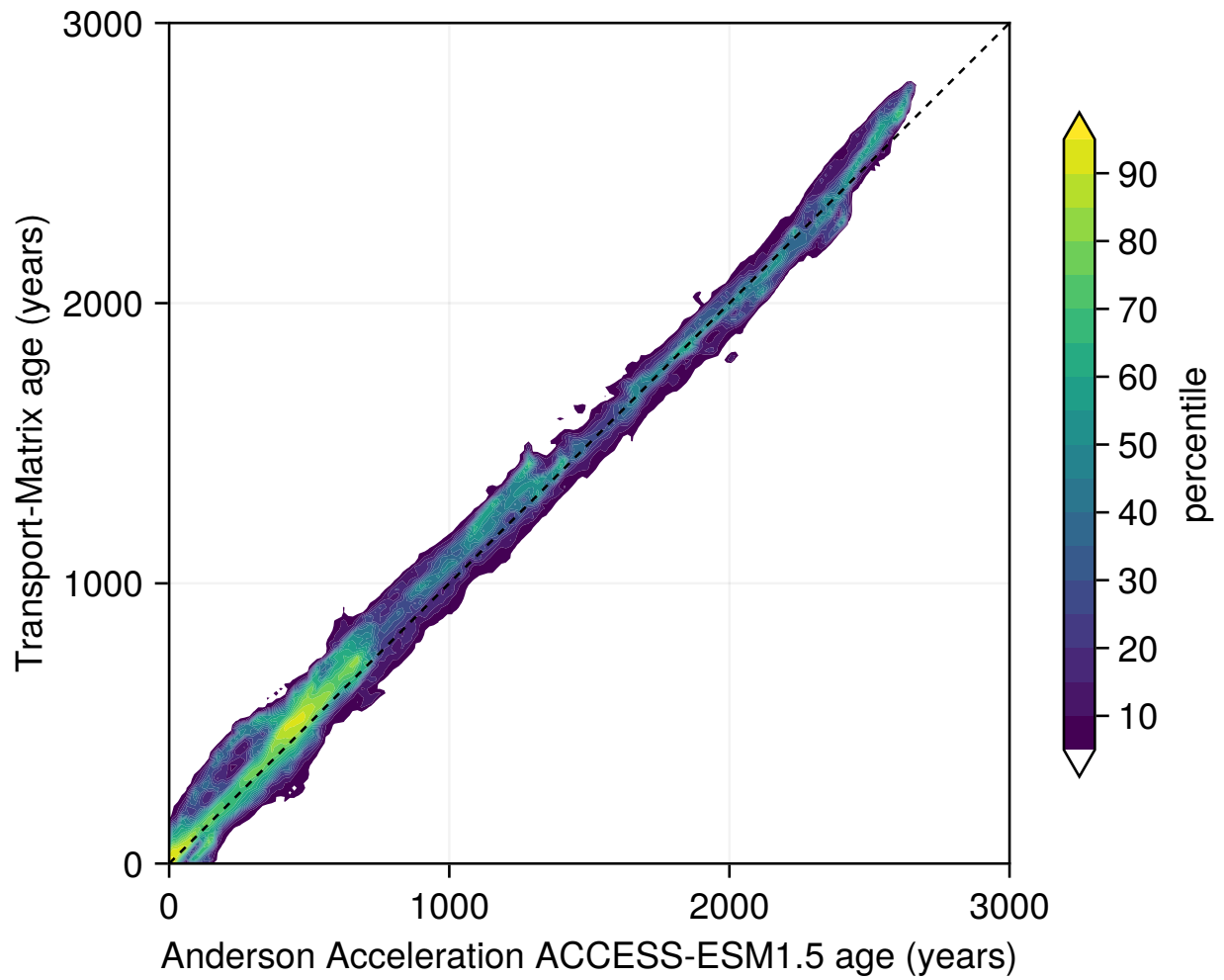


Figure S3. Joint probability density of the transport-matrix (TM) age vs the Anderson Acceleration (AA) age.

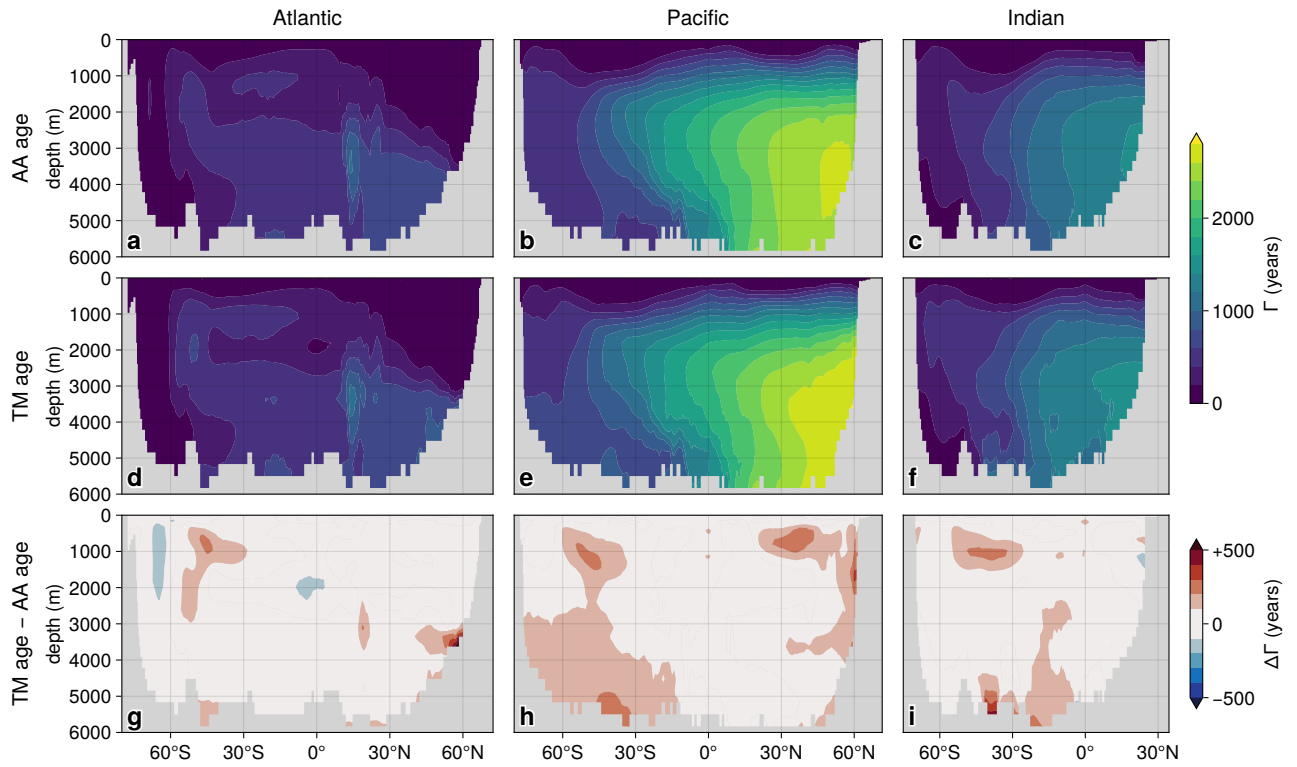


Figure S4. (a) Atlantic zonal mean of the ACCESS-ESM1.5 water age after Anderson Acceleration (AA). (b) As (a) but for the Pacific. (c) As (a) but for the Indian Ocean. (d–f) As (a–c) but for the transport matrix (TM) water age. (g–i) Difference between TM age (d–f) and AA age (a–c).

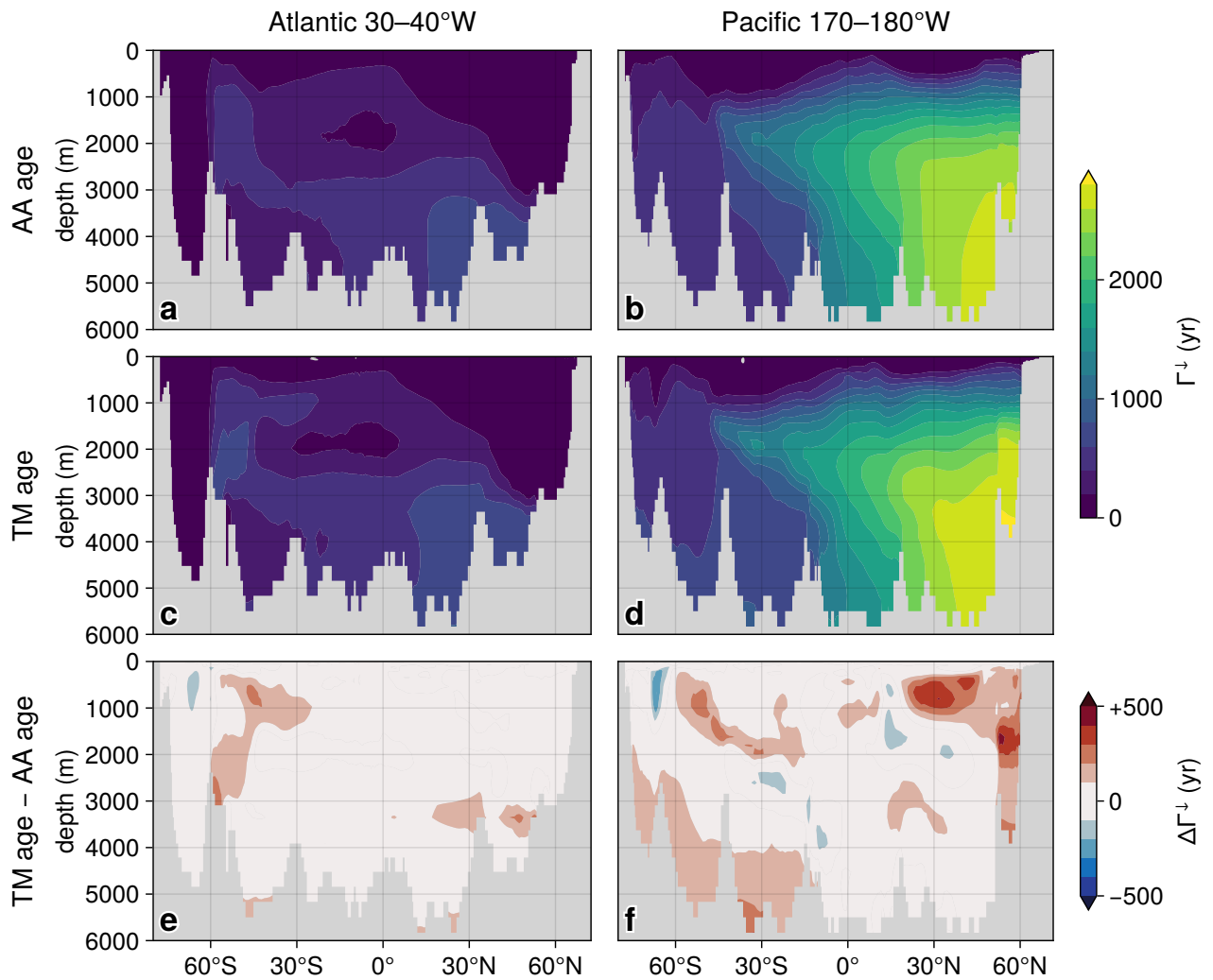


Figure S5. (a) 30–40°W Atlantic zonal mean of the ACCESS-ESM1.5 water age after Anderson Acceleration (AA). (b) As (a) but for the 170–180°W Pacific. (c–d) As (a–b) but for the transport matrix (TM) water age. (e–f) Difference between TM age (c–d) and AA age (a–b).

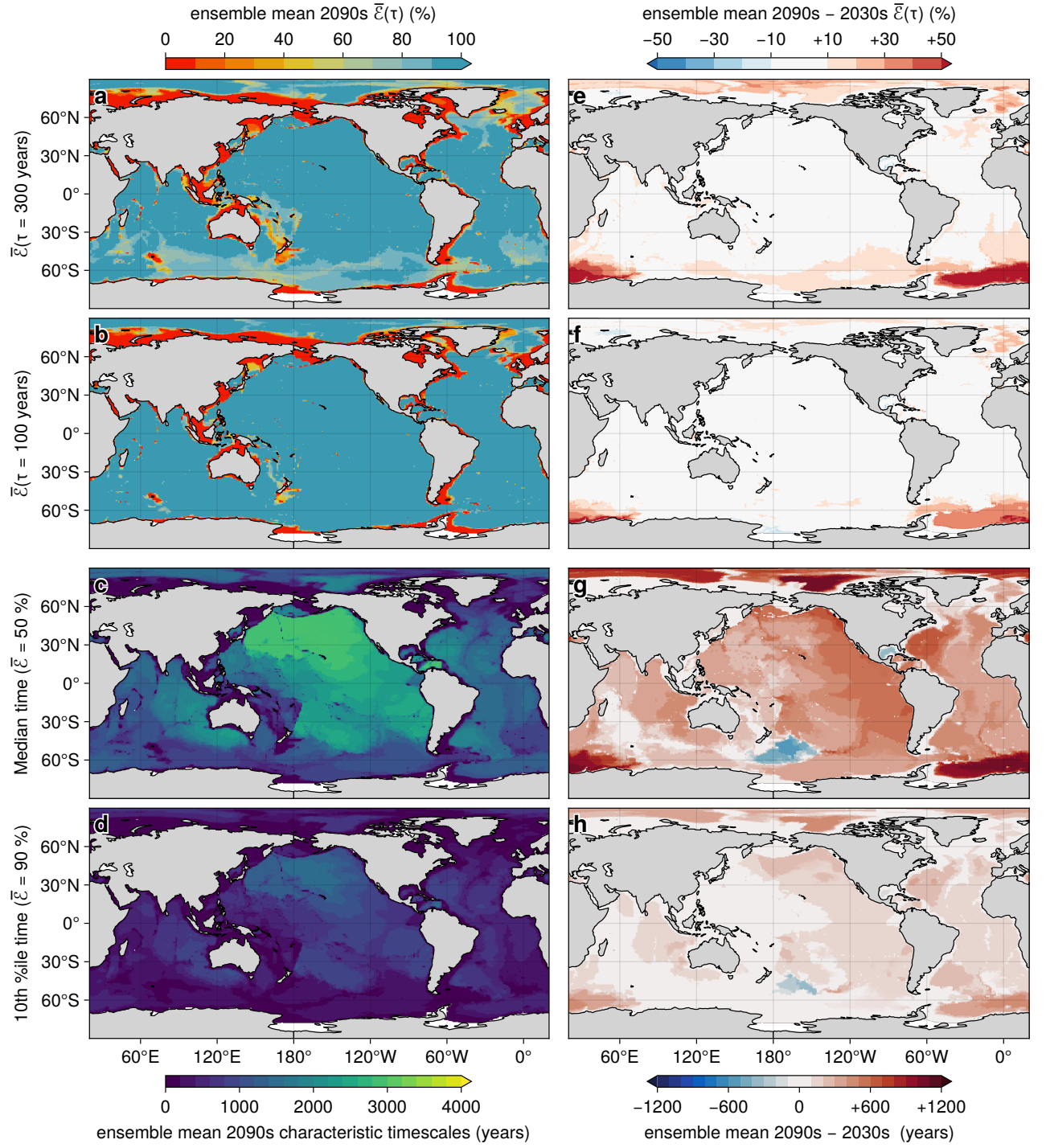


Figure S6. (a,b,e,f) as Fig. 4a,c but for the median and 10th percentile times. (c,d,g,h) as Fig. 4b,d but for $\tau = 100$ and 300 years.